

anchor-block-integrity-check

scripts/ anchor_block_integrity_check. sh — mechanical §11.4.227(B) anchor-block-integrity gate

Revision: 1 **Last modified:** 2026-08-19T00:00:00Z **Status:** active
Item: BOB-105 (§11.4.238 followup — mechanical §11.4.227(B)
anchor-block-integrity check)

Overview

scripts/anchor_block_integrity_check.sh mechanically enforces §11.4.227(B) — the rule that propagation-class gates count anchor BLOCK-STARTS across a declared lockstep mirror set, never bare literals. It resolves four defect classes that a ≥ 1 -literal presence-gate cannot see:

1. **BLIND-EXTRACTOR** — zero block-starts extracted from a listed file (§11.4.201(6) false-null guard).
2. **DUPLICATE-OR-COLLISION** — the same anchor id §11.4.NNN heads more than one block-start in a single file. A duplicate is usually a cut-and-paste mistake; a *collision* is §11.4.54's forbidden case — two different mandates minted under the same anchor number. Both fail with the same finding class; the block hashes tell them apart at review.
3. **LOCKSTEP-DIVERGENCE** — the same anchor's block has different content-hashes across the mirror set (§11.4.157 lockstep broken).
4. **LOCKSTEP-GAP** — the same anchor appears in some mirrors but not others (a mirror is missing the anchor entirely).

Sub-anchor prefix-match protection is built in: §11.4.10.A is a distinct id from §11.4.10, captured whole. Mid-body citations like §11.4.115(F) are excluded by construction (the block-start regex never accepts them at line start).

Configuration is DATA (§11.4.35)

The checker carries **no** project-specific literal. It reads its mirror-set + block-start regex from a config file resolved in this order:

- 1. `--config <path> argument`
- 2. `$ANCHOR_INTEGRITY_CONFIG env var`
- 3. `scripts/anchor_block_integrity_check.conf (default)`

The default config declares constitution/CLAUDE.md, AGENTS.md, QWEN.md, GEMINI.md as the byte-identical mirror set and Constitution.md as the canonical variant (excluded from mirror-hash equality per §11.4.227(B) canonical-vs-mirror layer variance).

Usage

```
# Run against the default mirror set:
scripts/anchor_block_integrity_check.sh

# Run against an alternate config (used by the challenge harness):
scripts/anchor_block_integrity_check.sh --config /path/to/
other.conf

# Verbose (per-file block-start count):
scripts/anchor_block_integrity_check.sh --verbose
```

Exit codes: - 0 — every invariant holds; PASS emitted to stdout. - 1 — one or more findings; each on stderr as FAIL: <class>: <detail>. - 2 — configuration/environment error (missing config, unreadable file). Distinct from finding-level FAIL per §11.4.201(4) conservative-safe default.

Anti-bluff — §11.4.107(10) self-validation

The checker ships with challenges/scripts/anchor_block_integrity_challenge.sh, which runs it against five fixtures under challenges/fixtures/anchor_block_integrity/:

fixture	expected rc	expected finding class
golden-good	0	(none)
negative-control	0	(none — mid-body citations)
golden-bad-duplicated	1	DUPLICATE-OR-COLLISION

fixture	expected rc	expected finding class
golden-bad-diverged	1	LOCKSTEP-DIVERGENCE
golden-bad-collision	1	DUPLICATE-OR-COLLISION

A checker that **PASS**es a golden-bad fixture, or **FAIL**s golden-good or negative-control, is itself the bluff (§11.4.107(10)) and the harness exits 1.

Findings from the first real run (2026-08-19)

The first run against the actual constitution/ mirror set reported **176 findings**, dominated by two classes:

- **LOCKSTEP-GAP** on the majority of §11.4.5..§11.4.99 range — anchors that carry a ### §11.4.NNN block-start in one mirror (typically CLAUDE.md or AGENTS.md) but no matching block-start in one or more of the other three mirrors. Per-mirror asymmetry is the mechanism §11.4.227(B) predicts: a ≥ 1 -literal presence-gate reads GREEN on each mirror in isolation while the mirror set as a whole is uneven.
- **LOCKSTEP-DIVERGENCE** on §11.4.69, §11.4.192, §11.4.193, §11.4.227 — the same anchor is block-started in every mirror but the block body differs, so the four mirrors are not byte-identical for those anchors.

These findings are the §11.4.238 discovery-channel-escape signal BOB-105 was minted to surface. Closure of individual findings is constitution-submodule work outside this boba-side checker's scope; this script's job is to make the class OBSERVABLE from a boba pre-build gate per §11.4.226 (evidence-class-at-closure).

Relationship to other anchors

- **§11.4.227(B)** — the rule this script mechanises.
- **§11.4.157** — content-hash lockstep across the mirror set.
- **§11.4.54** — anchor numbers never reused (collision detection).
- **§11.4.201(6)** — false-null guard on the extractor itself.
- **§11.4.107(10)** — golden-good + golden-bad + negative-control validation of the checker as an oracle.
- **§11.4.35** — consumer supplies mirror set + regex as DATA.
- **§11.4.6** — no guessing; every FAIL carries the resolved anchor id + file + hash prefix as captured evidence.

Companion files

- `scripts/anchor_block_integrity_check.sh` — the checker itself.
- `scripts/anchor_block_integrity_check.conf` — the default config for boba's own constitution mirror set.
- `challenges/scripts/anchor_block_integrity_challenge.sh` — the §11.4.107(10) harness.
- `challenges/fixtures/anchor_block_integrity/` — the five fixtures.